CLA 4 Report

ML Model Deployment Strategies and Model Serving using FastAPI

Student Name: _____ **Roll Number:** _____

Dataset: Titanic – Machine Learning from Disaster (Kaggle)

Objective

Train and evaluate a classification model, demonstrate online, batch, and streaming inference, and expose the trained pipeline through a FastAPI REST API.

Methodology

The Kaggle train.csv dataset is preprocessed using median imputation for numerical features, most-frequent imputation and one-hot encoding for categorical features. A Random Forest classifier is trained on a stratified training split and evaluated on a held-out test split. The complete pipeline is saved as model.pkl.

Evaluation

Actual test accuracy: Fill this in from outputs/evaluation.txt after running python train_model.py.

Inference Strategies

Online inference predicts one held-out passenger. Batch inference predicts 20 held-out records and saves batch_predictions.csv. Streaming inference processes 10 records sequentially with a one-second delay. FastAPI exposes GET /health and POST /predict, with request validation using Pydantic.

Observations

1. Online inference handles one record per request; batch inference processes multiple records together.
2. Streaming inference processes records sequentially as they arrive; batch inference processes a collected group.
3. FastAPI provides HTTP endpoints, Pydantic request validation, and interactive testing through Swagger UI.
4. Reusing the same saved pipeline ensures consistent preprocessing and model behavior and avoids repeated retraining.

Conclusion

This assignment demonstrates the machine-learning workflow from model training and evaluation to multiple inference strategies and REST API serving. Add the measured accuracy and screenshots from your own execution before submitting.